

Composition Pair B1.01 + B1.06 — The Enabling Pair: Instinct/Reasoning Separation Requires the Two-Layer DNA/Action Structure as Its Substrate Implementation, and the Two-Layer Structure Is Architecturally Motivated by the Requirement to Keep Governing Rules and Operational Records Architecturally Distinct

A derivation note from the Coordination Knowledge Substrate (CKS) pattern.

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its contribution is to formalize the architectural interdependence between two foundational commitments — the instinct/reasoning separation (B1.01) and the two-layer DNA/Action structure within every cell (B1.06) — as a named composition pair with a specific enabling relationship, mutual dependency, and co-presence requirement. This is the second note in Phase B4 of the CKS derivation series, following the integrating frame established in B4.01, and the first individual composition pair note in that series.

Abstract

The Coordination Knowledge Substrate (CKS) pattern's Paper 2 commits to two foundational architectural commitments whose relationship is more than parallel. B1.01 — the instinct/reasoning separation, which holds that a CKS-governed AI Self separates fast-pattern instinct (the LLM) from deliberate reasoning (the CKS substrate) into independently-evolving layers — is a conceptual commitment. Without a substrate architecture that gives each side of the separation a physical home, the commitment exists in architecture documents but not in operational substrate. B1.06 — the two-layer DNA/Action structure within every cell, which holds that each cell carries a DNA layer of stabilized orchestration content and an Action layer of recorded task instances — provides that substrate architecture. The DNA layer is the substrate home for reasoning-layer content; the Action layer is the substrate home for instinct-layer operational outputs. The enabling relationship runs in both directions. B1.06 enables B1.01 by providing the substrate implementation the conceptual separation requires; B1.01 motivates B1.06's specific two-layer design by explaining why the layers must be kept architecturally distinct rather than arbitrarily separated. This note

formalizes the pair, identifies its mutual dependencies, states the co-presence requirement and what fails when either commitment is present without the other, describes how the pair operates together in a governed substrate, and identifies the detection tests and failure modes that characterize partial presence.

1. Why formalizing this pair is architecturally necessary

Phase B4 of the CKS derivation series formalizes composition pairs: cases where two commitments are not merely co-present in the architecture but structurally interdependent in a way that makes each commitment load-bearing for the other. The B1.01 + B1.06 pair is the foundational instance of this relationship within Paper 2's architecture. It is the enabling pair from which the rest of the architecture's substrate structure follows.

The reason the pair requires separate formalization, rather than being derivable from the two commitments individually, is that neither commitment is fully characterizable without reference to the other. B1.01's instinct/reasoning separation specifies that the LLM and the CKS substrate are architecturally distinct, that the substrate carries governed reasoning content while the LLM carries fast-pattern instinct, and that the two layers evolve independently under different mechanisms. But the separation as stated in B1.01 does not itself specify where, within a cell's substrate, reasoning-layer content lives, nor where the operational outputs of the instinct layer are recorded. Without B1.06, B1.01 is a separation principle applied to the architecture as a whole, but not a substrate-level commitment about content location within cells. Similarly, B1.06's two-layer design — DNA as stabilized orchestration content, Action as recorded task instances — does not itself specify why the two layers must remain architecturally distinct rather than being merged into a unified cell substrate. The two-layer structure could in principle be treated as an organizational convenience rather than an architectural requirement. Without B1.01, there is no architectural reason why content should not flow freely between DNA and Action layers. B1.01 provides that reason: the DNA layer holds reasoning-layer content whose authority status must not be conflated with the instinct-layer outputs held in the Action layer.

The pair is therefore enabling in both directions, and both directions must be stated to fully characterize either commitment.

2. Pair identification

Commitment B1.01 — The instinct/reasoning separation. A CKS-governed AI Self separates fast-pattern instinct from deliberate reasoning into two independently-evolving layers under unified human governance (Paper 2, §4). The instinct layer is the LLM, operating as a System-1 analogue — pattern-matched, fast-path responses. The reasoning layer is the CKS substrate from Paper 1, operating as a System-2 analogue — explicit, deliberate, human-governed processing of what humans choose to encode. The two layers compose into one Self under unified human governance,

with the LLM's capability essential at every scope and the substrate providing the architecture that makes that capability work in coordinated, governed, evolving systems. The separation is governance-motivated, not capability-motivated: the reasoning layer carries what humans govern, the instinct layer carries what the model handles at speed, and the boundary between them is itself governed substrate content.

Commitment B1.06 — The two-layer DNA/Action structure within every cell. Within every cell, two architecturally distinct substrate layers operate (Paper 2, §5.3). The DNA layer holds stabilized orchestration substrates — the content that defines what a cell is governed to do: orchestration rules, behavior substrates, schemas, and authority policies. The Action layer holds recorded task instances — what the cell has actually done: outputs, execution records, and lived operational history. The two layers are substrate content under Paper 1's commitments in all cases; they are not functional layers in the computational sense but substrate content categories with distinct governance semantics.

Relationship type: Enabling pair with bidirectional motivation. B1.06 enables B1.01 by providing the substrate architecture that gives the conceptual separation its physical implementation. B1.01 motivates B1.06's specific two-layer design by providing the architectural reason those layers must be kept distinct. The pair is not unidirectional and not merely co-present: neither commitment is architecturally characterizable without the other.

3. Enabling direction: how B1.06 enables B1.01

B1.01's instinct/reasoning separation is a commitment about the architecture of a CKS-governed AI Self at the level of the whole system. The commitment specifies that two layers exist, that they are distinct, that they evolve independently, and that the substrate carries reasoning content while the LLM carries instinct. What B1.01 does not specify — and cannot specify at its own level of abstraction — is where within a cell's substrate the reasoning layer's content physically resides, and where the instinct layer's operational outputs are recorded when a cell executes work.

This is the architectural gap B1.06 closes. The DNA layer is the substrate location for reasoning-layer content at the cell level: the orchestration rules that govern what the cell does, the behavior substrates that encode governed deliberation, the schemas that structure what the LLM can read and what it can write. These are the content categories B1.01 assigns to the reasoning layer — explicit, governed, human-modifiable — and B1.06 gives them a named, architecturally distinct home within every cell. The Action layer is the substrate location for the outputs of instinct-layer operations at the cell level: recorded task instances, execution outputs, and operational history. These are the content categories that accumulate from LLM-mediated work — they represent what the instinct layer produced, under what rules, and at what time. B1.06 gives these a named, architecturally distinct home as well, separate from the DNA layer's governing content.

Without B1.06, the instinct/reasoning separation established in B1.01 has no substrate architecture within which to operate. The principle that the reasoning layer's content is authoritative and governed would have no physical substrate location; the principle that the instinct layer's outputs are recorded separately would have no substrate home. B1.01's separation can be expressed in architecture documentation without B1.06, but it cannot be tested, inspected, governed, or enforced at the substrate level. The reproducibility test — which verifies that the same inputs under the same DNA produce the same outputs — can only function if the DNA layer is a separately-maintained, stable substrate. Without B1.06's explicit DNA layer, there is no stable reasoning substrate to test against. B1.06 is the substrate implementation that makes B1.01 operational rather than merely nominal.

4. Motivating direction: how B1.01 motivates B1.06's specific design

B1.06's two-layer structure is not arbitrary. Any number of organizational schemes could divide cell substrate content into categories. The specific division into DNA (stabilized orchestration content) and Action (recorded task instances) is architecturally motivated by B1.01's instinct/reasoning separation, and the requirement that the layers remain architecturally distinct — rather than merely administratively organized — follows directly from B1.01's commitment about what conflation would cost.

The cost of conflation is precisely specified by B1.01: the reasoning layer carries governing rules whose authority is human-established and whose stability enables reproducible governance. The instinct layer's outputs in the Action layer carry no such authority — they are records of what the LLM produced under current governing rules, not themselves governing content. If DNA-layer content and Action-layer content are held in the same undifferentiated substrate location, two failure modes follow. First, governance reads of the reasoning layer become unreliable: a governance affordance that reads what rules apply to this cell cannot distinguish between a governing rule (which has authority) and a recorded output (which does not). Second, the Action layer's operational history can silently accumulate in ways that affect what the next governance read returns — the instinct layer's outputs can drift into the governing content without a governed transition. Both failure modes are instances of Layer Conflation (B3.10), the anti-pattern where the two-layer structure exists but the architectural motivation for keeping the layers distinct has been forgotten.

B1.01 provides the architectural reason why B1.06's layers must be distinct at the substrate level, not merely organized separately in documentation or by convention. The distinction is architecturally required because the two content types represent opposite sides of the governance boundary: reasoning-layer content is authoritative for behavior (DNA); instinct-layer output is a record of behavior performed under that authority (Action). Conflating them would conflate authority with its exercise — which is precisely the conflation B1.01's separation is designed to prevent.

The two-layer design of B1.06 is thus not a free-standing architectural decision. It is the substrate-level implementation of a governance distinction that B1.01 established at the conceptual level. A deployment that adopted B1.06's two-layer structure without understanding B1.01's motivating separation would have the structure but not the reason — and without the reason, the enforcement of the structural distinction becomes optional rather than architecturally required.

5. Mutual dependency

B1.01's dependency on B1.06. B1.01's instinct/reasoning separation commits to governed reasoning content that is stable, inspectable, and separately maintained from the instinct layer's operational outputs. This commitment requires a substrate architecture that physically separates the two content types within every cell. The reproducibility guarantee — that the same inputs under the same governing rules produce verifiable outputs — can only be tested if the governing rules reside in a stable, addressable substrate location distinct from operational records. Without B1.06's DNA layer as the stable reasoning substrate, there is no substrate location to address when verifying governing-rule stability. Without B1.06's Action layer as the record of instinct-layer outputs, there is no substrate basis for the action-feedback evolution mechanism that closes the loop between operational experience and DNA refinement. B1.01 requires B1.06 not as a downstream convenience but as the operational substrate without which B1.01's architectural commitments cannot be verified, tested, or governed at the cell level.

B1.06's dependency on B1.01. B1.06's two-layer structure commits to architectural distinctness between DNA and Action layers, not merely organizational separation. This architectural distinctness requires a principled reason for the distinction — a reason that tells governance implementers why content cannot freely migrate between layers and why the layers must be maintained under different governance processes. B1.01 provides that reason. The DNA layer must be distinct from the Action layer because DNA-layer content (reasoning-layer content) carries governance authority that Action-layer content (instinct-layer outputs) does not. Without B1.01's separation principle as the motivating rationale, B1.06's architectural requirement for distinctness becomes an arbitrary design convention that can be relaxed when operationally convenient. The Layer Conflation anti-pattern (B3.10) is the failure mode that results: layers exist in the structure but content bleeds between them because the separation's architectural motivation — B1.01 — is absent. B1.06 requires B1.01 not as theoretical background but as the load-bearing rationale that makes the layers' architectural distinctness enforceable rather than advisory.

6. Co-presence requirement

Full co-presence of B1.01 and B1.06 is architecturally required for either commitment to be fully operational. The two partial-presence cases each name a distinct failure mode.

B1.01 present, B1.06 absent — principle without substrate implementation. A deployment that commits to the instinct/reasoning separation (B1.01) but does not implement the two-layer

DNA/Action structure (B1.06) has the separation as an architectural principle but no substrate architecture within which to instantiate it. The reasoning layer's governing rules and the instinct layer's operational outputs exist somewhere in the cell's substrate, but there is no named, architecturally distinct location for either. The separation is expressed in documentation, but the substrate does not enforce it. Governance affordances cannot reliably distinguish governing content from operational records; reproducibility tests cannot address a stable DNA layer because none has been defined; action-feedback evolution cannot distinguish the source of the feedback signal from the governing content it is intended to refine. This failure mode is Instinct-Reasoning Collapse at the substrate level: the separation principle exists conceptually but lacks substrate implementation, and the architecture's downstream commitments that depend on the separation holding at the substrate level are unreliable. B3.02 in the anti-pattern series names the associated failure.

B1.06 present, B1.01 absent — structure without enforcement motivation. A deployment that implements the two-layer DNA/Action structure (B1.06) but does not hold B1.01's instinct/reasoning separation as the motivating principle has the structural distinction but lacks the architectural reason for maintaining it under operational pressure. Without B1.01's separation principle as the governing rationale, the architectural requirement for keeping the layers distinct is not grounded in any commitment the deployment is making about the authority difference between governing rules and operational records. Implementation teams encountering operational friction have no architectural argument against expedient choices that blur the layer boundary. The layers drift. Governing rules accumulate in the Action layer; Action-layer records pollute the DNA layer. The two-layer structure continues to exist nominally, but its architectural distinctness erodes. This is Layer Conflation (B3.10): the structure B1.06 defines is present, but the separation motivation B1.01 provides is absent, and the layers' content bleeds in ways that undermine both reproducibility and governance reliability.

Full co-presence. When both B1.01 and B1.06 are present, the separation is both principled (B1.01 provides the governance-motivated reason for the distinction) and implemented (B1.06 provides the substrate architecture in which the distinction is enforced). Governance affordances can inspect, test, and enforce the separation through the substrate: the DNA layer is addressable as the reasoning substrate; the Action layer is addressable as the operational record; the boundary between them is architecturally maintained because both the structural distinction (B1.06) and its motivating rationale (B1.01) are operative. The reproducibility test functions because the DNA layer is a stable, separately-maintained substrate. Action-feedback evolution operates correctly because the feedback signal (Action layer) is architecturally distinct from the content it is intended to refine (DNA layer), and the governed transition between them is the only path by which operational experience enters governing content.

7. Operational integration

The pair operates as follows in a governed substrate.

The LLM performs operational work as the instinct layer per B1.01. Its outputs — task results, responses, execution artifacts — are recorded in the Action layer per B1.06. The Action layer is the substrate record of what the instinct layer produced; it accumulates over the cell's operational lifetime and is the source material for action-feedback evolution. The LLM does not write to the DNA layer directly: instinct-layer outputs are not self-authorizing as governing content, and the architectural boundary B1.06 maintains between the two layers — motivated by B1.01's separation — enforces this.

The reasoning layer provides governing rules as the CKS substrate per B1.01. Those rules reside in the DNA layer per B1.06. When the LLM consults the reasoning layer during AI-as-substrate-mediation (per Paper 1, §4.2), it reads DNA-layer content — the orchestration rules and behavior substrates that govern what it is to do in this cell. The DNA layer is the substrate home for the reasoning layer's authority at the cell level. It is human-maintained, addressable, and stable across the cell's operational lifetime unless a governed DNA evolution event updates it.

Governance affordances operate on DNA-layer content — they read, inspect, modify, and authorize changes to reasoning-layer governing content — without modifying Action-layer content (instinct records). This governance isolation, implemented by B1.06's layer distinction, is the substrate-level expression of B1.01's requirement that the reasoning layer be separately maintainable and separately evolvable from the instinct layer's operational outputs.

Action-feedback evolution — when operational experience accumulated in the Action layer informs a refinement of DNA-layer content — crosses the layer boundary under governed transition: a human authority reviews the Action-layer signal and authors a change to the DNA layer. The transition is governed, not automatic, because the movement of information from instinct-layer records to reasoning-layer governing content is precisely the kind of change B1.01's separation identifies as requiring human governance. B1.06's architectural layer distinction is what makes the transition detectable as a cross-boundary event requiring governance, rather than an invisible content drift.

8. Detection and failure modes

Detection tests for correct pair composition.

B1.01's separation principle is operative: Ask whether the deployment's governance documentation can state, for each category of substrate content, whether that content belongs to the reasoning layer (DNA) or the instinct layer's operational record (Action). If the answer requires inspection of the actual substrate rather than derivation from a governing rule, B1.01's separation principle is not operative as an architectural commitment.

B1.06's two-layer structure is architecturally implemented: Ask whether the cell's substrate has a named, addressable DNA layer and a named, addressable Action layer. Addressability means that a governance read can retrieve DNA-layer content without retrieving Action-layer content, and vice

versa. If the cell's substrate holds all content in an undifferentiated store from which layer membership must be inferred by content inspection, B1.06 is not architecturally implemented.

Pair completeness: Ask whether B1.01's separation principle is governing how the two layers are maintained — specifically, whether the architectural reason given for the DNA/Action distinction is the authority difference between governing rules and operational records. If the two-layer structure is maintained by convention rather than by a stated architectural reason, B1.01's motivating principle is not informing B1.06's enforcement, and the pair is not fully composed.

Failure mode: Layer Conflation (B3.10). The failure mode when B1.06 is present without B1.01's motivating separation. Symptoms include: governing rules added directly to the Action layer without a governed DNA evolution event; operational outputs in the DNA layer treated as having governing authority; governance reads of the DNA layer that return Action-layer content; action-feedback evolution that modifies DNA-layer content without a governed transition. The architectural correction is to reinstate B1.01's separation principle as the governing rationale for B1.06's layer distinction.

Failure mode: Instinct-Reasoning Collapse (B3.02). The failure mode when B1.01's separation principle is present without B1.06's substrate implementation. Symptoms include: no addressable DNA layer distinct from the general cell substrate; governing rules and operational records held in the same substrate location; reproducibility tests that cannot address a stable DNA layer; action-feedback evolution that cannot distinguish the feedback signal from the governing content it should refine. The architectural correction is to implement B1.06's two-layer structure so that B1.01's separation principle has a substrate architecture in which to operate.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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